

e2682995.pdf

Emine Gökçe Kekilli

May 2026

Part 1: NAND/NOR-Based D Latch Design

Questions

1. Why is the D latch called level-triggered? Please use the behavior of EN in your explanation.

Answer: The D latch's operation behaves according to EN input, not clock input. This circuit is called level-triggered because it operates not just for an instant, but over a specific level of the EN. While $EN = 1$, the latch is transparent. Q tracks D continuously. While $EN = 0$, the latch holds the previous value of Q.

2. While $EN=1$, does the output change at exactly the same instant as the input? If not, what physical effect causes the difference?

Answer: No, the output does not change at exactly the same instant as the input. Physical propagation delay causes this difference. Logic gates need a finite amount of time to switch states because of their internal structures, this emerges propagation delay.

3. What is the forbidden or contradictory condition of the underlying SR core, and how does your D-latch design prevent it?

Answer: The forbidden or contradictory position of the underlying SR latch occurs when $S=0$ and $R=0$ (NAND-based). This forces both Q and $\bar{Q}$ to 1. The D-latch design uses a NOT gate to ensure that the set and reset inputs always receive opposite logic levels. Thus, S and R never be active at the same time and the forbidden case does not happen.

4. If D changes just before the latch closes, what value do you expect the latch to store? Please explain using the timing of the enable transition.

Answer: The latch is expected to store the value of D that was present at the exact moment of the logic value of EN reduces from 1 to 0. As long as the change in D satisfies the required setup time before the latch closes, the new value will be kept and maintained when the EN reaches the low level.

Part 2: Master-Slave Positive-Edge-Triggered D Flip-Flop

1. Please explain, using one concrete input scenario, the difference between the level-triggered D latch of Part 1 and the edge-triggered D flip-flop of Part 2.

Answer: We can give an example: If we change logic value of D from 0 to 1 while CLK=1 EN=1 for the latch, the D-latch output (Q) will change to 1. However, the D flip-flop output will not change because it only captures data at the rising edge (0 → 1) of the clock; since the clock is already 1, the change does not happen in D flip-flop.

2. Why must the master and slave not be active at the same time? Please explain race-through in your own words.

Answer: The master and slave latches must never be active simultaneously to ensure that the flip-flop operates strictly as an edge-triggered device. In a positive-edge-triggered D flip-flop, the master latch is active when CLK=0, and the slave latch is active when CLK=1. If both were active at the same time, "race-through" would happen. In this scenario, the input signal D would "race through" both latches and reach the output Q immediately without being held or sampled by the clock edges. This would make the flip-flop behave like a simple transparent latch. Therefore we guarantee that data is captured only at the exact instant of the rising edge.

3. If your manual clock source bounces, how can that distort the expected behavior? What symptoms would you expect at the output?

Answer: In a clock bouncing situation there are multiple unintended rapid transitions between 0 and 1. This can cause the flip-flop to trigger several times in a single press. We can see incorrect data captures, unstable output and unexpected state changes.

Part 3: D Flip-Flop with Asynchronous Preset and Clear

Questions and Answers

1. What is the main difference between synchronous and asynchronous signals?

Answer: The main difference is related to the clock signal. Asynchronous signals operate independent of clock signal. For example, the asynchronous preset changes the output immediately without waiting for the clock signal. Synchronous signals depend on clock signal. They cannot work without clock signal. For example, even if preset is 1, the circuit waits for the clock signal.

2. If $\overline{\text{PRE}}$ is asserted (and CLR is 0) while $D=0$ and the clock is switching, why must the output still go to 1?

Answer: Because asynchronous signals have higher priority than the clock input and the D input. PRE is an asynchronous input, so it changes the output with higher priority.

3. Please describe a scenario where the preset input is handy for a D-Flip-Flop.

Answer: A preset input is useful when a circuit needs to start with $Q=1$. For example, in an alarm system, the alarm may need to be active when the system be active. The preset input can makes the D Flip-Flop output 1 immediately without waiting for a clock signal.

4. Please describe a scenario where the clear input is handy for a D-Flip-Flop.

Answer: A clear input is useful to reset the output to 0 quickly. For example, in a counter circuit, the system may want to reset all outputs to 0. The clear input makes the D Flip-Flop output to 0 without waiting for the clock signal.

Part 4: JK Flip-Flop Implementation

Questions and Answers

1. Please write the behavior equation of the J-K Flip-Flop, including the D-Flip Flop inputs and outputs.

Answer: $Q_{\text{next}} = JQ' + K'Q$ When we use D Flip-Flop $D = JQ' + K'Q$
 Q is the current output Q_{next} : the next output J and K : inputs of the J-K Flip-Flop D : the input of the D Flip-Flop

2. How many additional gates did Part 4 require beyond Part 3?

Answer: 2 AND Gates: One for the JQ' and one for the $K'Q$. 1 OR Gate: $JQ' + K'Q$. 1 NOT Gate: For inverter of K

3. Why is the JK front-end logically richer than simply wiring D directly, and what extra timing sensitivity can that introduce in practice?

Answer: A JK flip-flop is more powerful than a D flip-flop because it can do some operations like set, reset, hold, or toggle. It uses J , K , and the current output Q to decide the next state. This extra logic makes it more flexible, but also more sensitive to timing. We can give an example for this situation: If $J = K = 1$ and the clock is high for too long, the output may change many times because of feedback delay.